



COMPUTER SCIENCE AND SOCIETY

AGENDA

Computer Science, Society, and the Information Society

1

The Role of Data in the Information Society

2

Economic Impact of the Information Society

3

Social Impacts of the Information Society

4

Infrastructure Vulnerability, Computer Science and the Military

5

Responsibility in Computer Science

6

UNIT 6

RESPONSIBILITY IN COMPUTER SCIENCE



On completion of this unit, you will be able to...

- explain the importance of **ethical aspects** in computer science.
- name important **ethical principles**.
- name the role of ethics in **algorithmic decision-making**.
- recommend mechanisms for **implementing responsibility** in organizations.



1. Explain what is meant by **Ethics**.
2. Explain ethical issues in **algorithmic decision making**.
3. Explain why we needed a **new kind of ethics** in the 20th century.

Why does it matter?

- Inventions affect all of humanity: e.g. **Internet**, **nuclear bomb**, ...
- Many potential issues, e.g.:
 - **Jobs lost** due to AI
 - Monoploies of large Internet Corporations holding huge amounts of **personal data**
 - Is it ethical to work for a company **selling IT to the military** (which one)?
 - ...

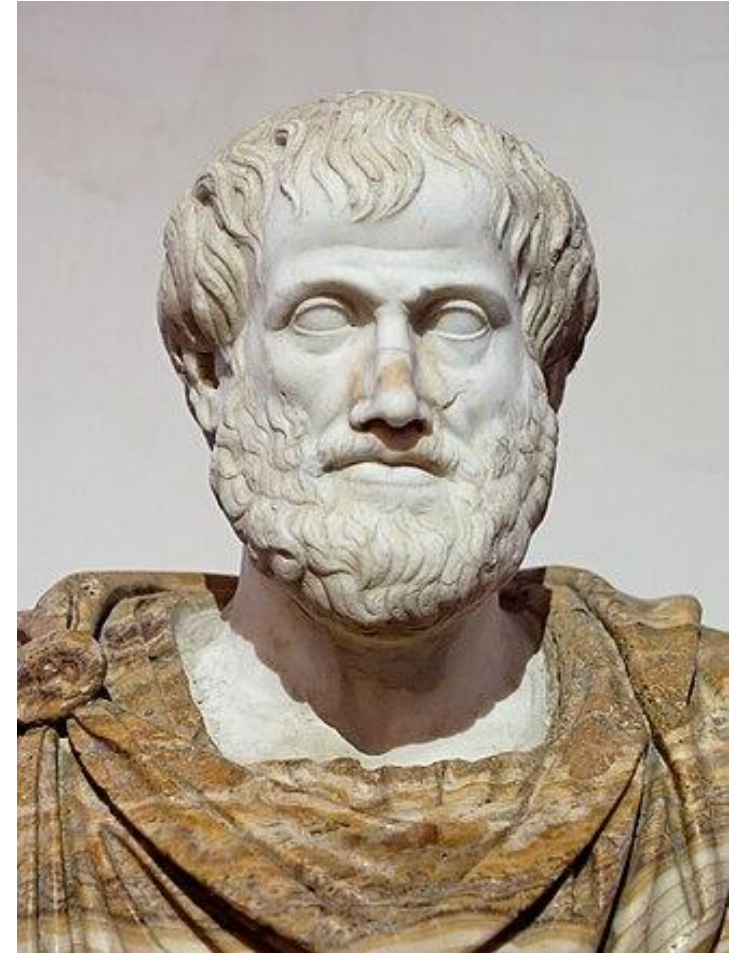


ETHICS

How should we act?

Moral actions: based on a group's **values and norms**

- Aristotle (384-322 BC): Nicomachean Ethics
- > foundation of **European culture**
- > also: historically/ **socially shaped**



IMMANUEL KANT (1724-1804)

Categorical imperative: **"Act only according to that maxim you can at the same time will as a universal law without contradiction."**

→ What would the world be like if your actions were turned into a **general law**?



- Technology has changed the world → actions of an individual can affect **all of humanity**
- New ethics needed
- “Act in such a way that the effects of your action are compatible with **the permanence of genuine human life on earth**”



Encourage ethical behavior, e.g. guidelines of GI/
German Informatics Society:

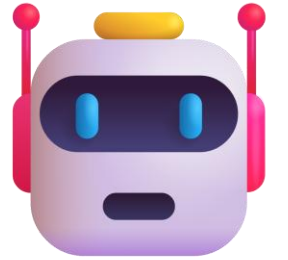
- matters of professional ethics or moral conflicts should be the subject of **collaborative reflection and action**
- **critically scrutinize** actions in social discourse
- call for **civil courage** that also justifies public whistleblowing
- “there can be **no conclusive instructions** or rigid rules”

GESELLSCHAFT
FÜR INFORMATIK



Decisions made automatically by algorithms:

- E.g. Human Resources (which candidate to hire?)

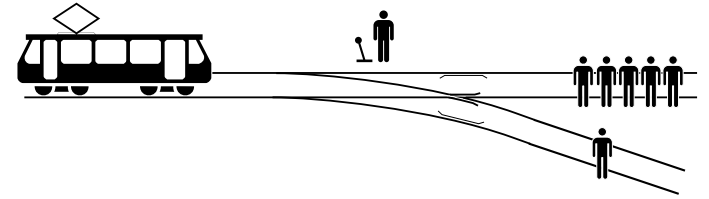


Issues:

- Bias?
- Lack of transparency?

THE NEW TROLLEY PROBLEM

- 1930s: driverless trolley rolls toward **five people** – pulling a lever would save them, killing only **one other person**
- What to do? **no universal answer**, ethical decision
- Today: humans write code for **self-driving cars** which may face similar issues



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IMPLEMENTING RESPONSIBILITY

- **Standards** can be defined for an organization or company, which then only need to be discussed when the framework conditions change.
- E.g. Google, “**Don’t be Evil**”
- Recommendations for **codes of conduct** from organizations like the Association for Computing Machinery (ACM)
- **Corporate (social) responsibility**: rules set internally by companies, respecting needs of their stakeholders



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SESSION 3

TRANSFER TASK

Floridi (2023) proposed a framework of five **principles for ethical AI**:

- 1. Beneficence:** Promoting Well-Being, Preserving Dignity, and Sustaining the Planet
- 2. Nonmaleficence:** Privacy, Security, and ‘Capability Caution’ (responsibility for risks arising from technological innovations)
- 3. Autonomy:** The Power to ‘Decide to Decide’ (instead of letting algorithms decide)
- 4. Justice:** Promoting Prosperity, Preserving Solidarity, Avoiding Unfairness
- 5. Explicability:** Enabling the Other Principles through Intelligibility and Accountability

TRANSFER TASK

1. In your group, discuss whether these guidelines make sense for you. **What risks do you see** in AI? Would the guidelines address these?
2. Apply the guidelines to a **scenario** of your choice (e.g. implementing an AI-based algorithm in a work process, regulating safe and ethical use of ChatGPT and similar tools in your organization, ...).
 - a. How well do they fit?
 - b. Turn the general guidelines into **specific guidelines** for your scenario!

TRANSFER TASK
PRESENTATION OF THE RESULTS

Please present your
results.

The results will be
discussed in plenary.



Sample solution

- 1. Risks** in AI could include overreliance on inaccurate tools, spread of disinformation, loss of jobs, ... Hallucinations (factually wrong answers) could also be seen as a risk. The guidelines address them well, but are very general. E.g. does “promoting well being” mean “do not destroy jobs”? Are intelligibility and accountability usually given in tools like ChatGPT, or are they black boxes?

Sample solution

2. Applying the guidelines to a **scenario** shows how they can be useful in practice.
- When implementing an AI-based algorithm in a work process: Let'S say a company automates its recutting process. Is the automated process fair (or is there bias)? Is it intelligible (or a black box?) Do humans have the **Power to 'Decide to Decide'** in cases of doubt? Guidelines could stipulate that criteria should be transparent and fair and that decisions can be audited. Also, who is responsible in case of bad decisions?
 - For safe and ethical use of ChatGPT and similar tools: avoid unnecessary use (saving energy), make sure the tool supports humans rather than supplanting them (solidarity), again, ensure decisions are fair and transparent
 - Of course, many other aspects are possible.



1. Who is regarded as one of the **first philosophers** to write about **ethics**?

- a) Aristotle
- b) Socrates
- c) Plato
- d) Immanuel Kant



2. What is not mentioned in the GI/ German Informatics Society guidelines?

- a) Members should **critically scrutinize** their actions
- b) Members should become **whistleblowers** if necessary
- c) Decisions should be made **collaboratively**
- d) Instructions are **conclusive**



3. What is a possible issue in **algorithmic decisions**?
- a) Cost
 - b) Bias
 - c) Quality
 - d) Time

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